

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 8

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Title

Implementation of HTTP Methods using JSON Server

Aim

To implement and demonstrate the **GET, POST, PUT and DELETE HTTP methods** using JSON Server for managing data in a RESTful web application.

Objectives

1. To understand the concept of a REST API and JSON Server.
2. To create a simple JSON-based REST API.
3. To perform **GET, POST, PUT and DELETE** operations.
4. To understand how client applications communicate with a server using HTTP methods.
5. To test API endpoints using a suitable tool such as **Postman** or a web browser.

Theory

JSON Server is a lightweight tool that allows developers to create a REST API from a JSON file without developing a complete backend server.

It uses HTTP methods to perform operations on resources:

HTTP Method	Purpose	Example
GET	Retrieve data	Get all students

POST	Add new data	Add a student
PUT	Update existing data	Update student details
DELETE	Remove data	Delete a student

For this experiment, a **db.json** file is used as the database.

Example resource:

```
{
  "students": [
    {
      "id": 1,
      "name": "Ravi",
      "branch": "CSE",
      "marks": 85
    }
  ]
}
```

When JSON Server is started, it automatically provides REST endpoints such as:

```
GET      /students
GET      /students/1
POST     /students
PUT      /students/1
DELETE   /students/1
```

Thus, JSON Server helps students understand RESTful API operations before working with a full backend framework.

index.html :

```
<!DOCTYPE html>
<html lang="en">
```

```
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-
scale=1.0">

  <title>Student Management System</title>

<style>
  body {
    font-family: Arial, sans-serif;
    margin: 40px;
    background-color: #f4f4f4;
  }

  h1 {
    text-align: center;
  }

  .form-container {
    background: white;
    padding: 20px;
    margin-bottom: 25px;
    border-radius: 8px;
  }

  input {
    padding: 10px;
    margin: 5px;
  }

  button {
    padding: 9px 14px;
    margin: 5px;
    cursor: pointer;
  }
```

```
table {
  width: 100%;
  background: white;
  border-collapse: collapse;
}

th,
td {
  padding: 12px;
  border: 1px solid #ccc;
  text-align: center;
}

th {
  background-color: #333;
  color: white;
}
</style>
</head>

<body>

<h1>Student Management System</h1>

<div class="form-container">

  <h2>Add Student</h2>

  <input type="text" id="name" placeholder="Student Name">

  <input type="text" id="branch" placeholder="Branch">

  <input type="number" id="marks" placeholder="Marks">

  <button onclick="addStudent()">Add Student</button>

  <button onclick="getStudents()">Refresh</button>

</div>
```

```
<div class="form-container">

  <h2>Student Records</h2>

  <table>

    <thead>
      <tr>
        <th>ID</th>
        <th>Name</th>
        <th>Branch</th>
        <th>Marks</th>
        <th>Actions</th>
      </tr>
    </thead>

    <tbody id="studentTable">
    </tbody>

  </table>

</div>

<script src="script.js"></script>

</body>

</html>
```

script.js

```
const API_URL = "http://localhost:3000/students";
```

```
// GET: Display all students
```

```
async function getStudents() {
```

```
  try {
```

```
    const response = await fetch(API_URL);
```

```

const students = await response.json();

const table = document.getElementById("studentTable");

table.innerHTML = "";

students.forEach(student => {

    const row = document.createElement("tr");

    row.innerHTML = `
        <td>${student.id}</td>
        <td>${student.name}</td>
        <td>${student.branch}</td>
        <td>${student.marks}</td>

        <td>
            <button onclick="updateStudent('${student.id}')">
                Update
            </button>

            <button onclick="deleteStudent('${student.id}')">
                Delete
            </button>
        </td>
    `;

    table.appendChild(row);
});

} catch (error) {

    console.error("Error:", error);

    alert("Unable to fetch student records.");

}
}

```

```
// POST: Add a new student
async function addStudent() {

    const name = document.getElementById("name").value;
    const branch = document.getElementById("branch").value;
    const marks = document.getElementById("marks").value;

    if (name === "" || branch === "" || marks === "") {

        alert("Please enter all student details.");

        return;
    }

    const student = {

        name: name,
        branch: branch,
        marks: Number(marks)

    };

    try {

        const response = await fetch(API_URL, {

            method: "POST",

            headers: {
                "Content-Type": "application/json"
            },

            body: JSON.stringify(student)

        });

        if (response.ok) {

            alert("Student added successfully.");
        }
    }
}
```

```

        document.getElementById("name").value = "";
        document.getElementById("branch").value = "";
        document.getElementById("marks").value = "";

        getStudents();

    }

} catch (error) {

    console.error("Error:", error);

    alert("Unable to add student.");

}

}

// PUT: Update student
async function updateStudent(id) {

    try {

        // Get current student details
        const response = await fetch(`${API_URL}/${id}`);

        const student = await response.json();

        // Ask for updated values
        const name = prompt("Enter student name:", student.name);

        if (name === null) {
            return;
        }

        const branch = prompt("Enter branch:", student.branch);

        if (branch === null) {
            return;
        }
    }
}

```



```
const marks = prompt("Enter marks:", student.marks);

if (marks === null) {
    return;
}

const updatedStudent = {

    name: name,
    branch: branch,
    marks: Number(marks)

};

// Update using PUT
const updateResponse = await fetch(`${API_URL}/${id}`, {

    method: "PUT",

    headers: {
        "Content-Type": "application/json"
    },

    body: JSON.stringify(updatedStudent)

});

if (updateResponse.ok) {

    alert("Student updated successfully.");

    getStudents();

}

} catch (error) {

    console.error("Error:", error);

}
```

```

        alert("Unable to update student.");
    }
}

// DELETE: Delete student
async function deleteStudent(id) {

    const confirmDelete = confirm(
        "Are you sure you want to delete this student?"
    );

    if (!confirmDelete) {
        return;
    }

    try {

        const response = await fetch(`${API_URL}/${id}`, {

            method: "DELETE"

        });

        if (response.ok) {

            alert("Student deleted successfully.");

            getStudents();

        }

    } catch (error) {

        console.error("Error:", error);

        alert("Unable to delete student.");

    }
}

```

```
}
```

```
// Load students when page opens  
getStudents ();
```

OUTPUT:

The image displays four sequential screenshots of a web browser at `http://localhost:5500/index.html`, illustrating the functionality of a Student Management System.

Screenshot 1 (Top Left): Shows the initial state with three students in the 'Student Records' table.

ID	Name	Branch	Marks	Actions
1	Ravi	CSE	85	Update Delete
2	Priya	AIML	91	Update Delete
3	Kiran	CSE	78	Update Delete

Output 1: GET request displays all student records.

Screenshot 2 (Top Right): Shows the result of a POST request to add a new student, Anil.

ID	Name	Branch	Marks	Actions
1	Ravi	CSE	85	Update Delete
2	Priya	AIML	91	Update Delete
3	Kiran	CSE	78	Update Delete
4	Anil	CSE	88	Update Delete

Output 2: POST request successfully adds a new student record.

Screenshot 3 (Bottom Left): Shows the result of a PUT request to update Ravi's marks to 90.

ID	Name	Branch	Marks	Actions
1	Ravi Kumar	CSE	90	Update Delete
2	Priya	AIML	91	Update Delete
3	Kiran	CSE	78	Update Delete
4	Anil	CSE	88	Update Delete

Output 3: PUT request successfully updates Ravi's details.

Screenshot 4 (Bottom Right): Shows the result of a DELETE request to remove student ID 3 (Kiran).

ID	Name	Branch	Marks	Actions
1	Ravi Kumar	CSE	90	Update Delete
2	Priya	AIML	91	Update Delete
4	Anil	CSE	88	Update Delete

Output 4: DELETE request successfully removes student ID 3.

Algorithm / Procedure

Step 1

Install Node.js and npm on the system.

Step 2

Create a new project folder named:

JSONServerDemo

Step 3

Open the terminal in the project folder and initialize the project:

```
npm init -y
```

Step 4

Install JSON Server: `npm install json-server`

Step 5

Create a file named: `db.json`

Step 6

Add sample student data to `db.json`

Step 7

Start JSON Server using: `npm run json-server --watch db.json --port 3000`

Step 8

Test the **GET** operation to retrieve all students.

Step 9

Test the **POST** operation to add a new student.

Step 10

Test the **PUT** operation to modify an existing student's details.

Step 11

Test the **DELETE** operation to remove a student.

Step 12

Verify the changes in `db.json` and record the outputs.

1. db.json

Create the following file:

```
{
  "students": [
    {
      "id": "1",
      "name": "Ravi",
      "branch": "CSE",
      "marks": 85
    },
    {
      "id": "2",
      "name": "Priya",
      "branch": "AIML",
      "marks": 91
    },
    {
      "id": "3",
      "name": "Kiran",
      "branch": "CSE",
      "marks": 78
    }
  ]
}
```

2. Start JSON Server

```
npx json-server --watch db.json --port 3000
```

```
PS C:\Users\Admin\OneDrive\Desktop\front-end classes> npx json-server --watch db.json --port 3000
--watch/-w can be omitted, JSON Server 1+ watches for file changes by default
JSON Server started on PORT :3000
Press CTRL-C to stop
Watching db.json...
```

(~ ~ ~)

Index:

<http://localhost:3000/>

Static files:

Serving ./public directory if it exists

Endpoints:

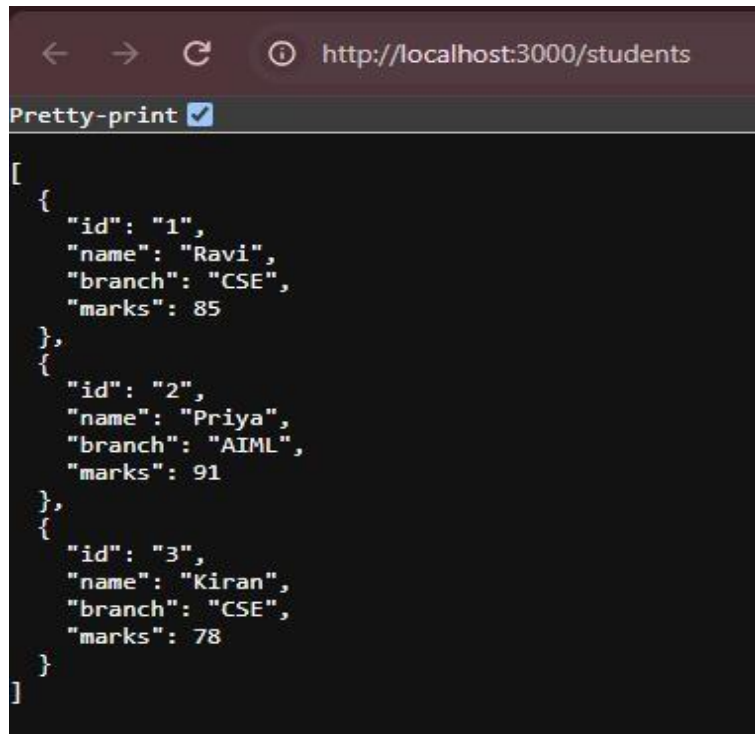
<http://localhost:3000/students>

HTTP Operations

A. GET Method

Retrieve all students

Method: GET



The screenshot shows a web browser window with the address bar displaying `http://localhost:3000/students`. The page content is a JSON array of three student objects, formatted with 'Pretty-print' enabled. The JSON data is as follows:

```
[
  {
    "id": "1",
    "name": "Ravi",
    "branch": "CSE",
    "marks": 85
  },
  {
    "id": "2",
    "name": "Priya",
    "branch": "AIML",
    "marks": 91
  },
  {
    "id": "3",
    "name": "Kiran",
    "branch": "CSE",
    "marks": 78
  }
]
```

B. POST Method

Add a new student

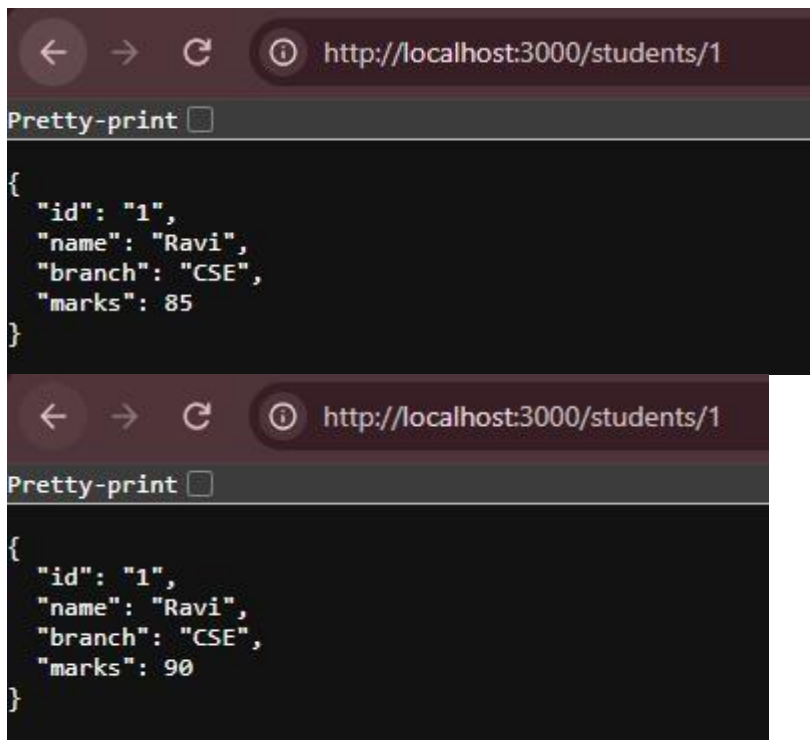
Method: POST

```
{  
  "name": "Anil",  
  "branch": "CSE",  
  "marks": 88  
}
```

C. PUT Method

Update student details

Method: PUT



The details of student with ID **1** will be updated.

D. DELETE Method

Delete a student

Method: DELETE

<http://localhost:3000/students/3> - The student with ID 3 will be removed.

Test Cases and Execution Data

Students should record at least the following test cases:

Test Case	HTTP Method	URL	Input / Operation	Expected Result
TC01	GET	/students	Retrieve all students	All student records are displayed
TC02	GET	/students/1	Retrieve student ID 1	Student ID 1 details are displayed
TC03	POST	/students	Add Anil, CSE, 88	New student record is created
TC04	PUT	/students/1	Change Ravi's marks to 90	Student ID 1 is updated
TC05	DELETE	/students/3	Delete student ID 3	Student ID 3 is deleted

Execution Data

Operation	Input	Actual Output	Status
GET all students	<code>/students</code>	3 student records displayed	Pass
GET student	<code>/students/1</code>	Ravi's details displayed	Pass
POST student	Anil, CSE, 88	New record added	Pass
PUT student	ID 1, marks = 90	Record updated	Pass
DELETE student	ID 3	Record deleted	Pass

Result

Thus, the GET, POST, PUT and DELETE HTTP methods were successfully implemented and tested using JSON Server for managing student records.